Hi all,

Sorry for the delay, I attached a short R lab that goes over how to fit the two competing risks regression models I talked about. This lab was compiled by Zian Zhuang, who is a current PhD student at UCLA under my PhD advisor. Most of the lab was actually from when I TA'd the survival analysis class when I was a PhD student.

In brief, the coxph function from the survival package fits a cause-specific Cox proportional hazards model. Note that you have to specify the cause/event of interest in the Surv() function. The crr function from the cmprsk package fits the Fine-Gray model, which models subdistribution hazards. Note that for the crr function you have to manually input a covariate matrix X, which can be a bit more tedious.

The coefficient estimates for both models give you log-HRs, exponentiating them gives you their respective hazard ratios.

For now, I'd start off with a simple model that only includes escalation status to see how the unadjusted results look. You should have 4 HRs, 2 events x 2 models). Afterwards, you can consider a multivariable model adjusting for the following confounders: Age at listing, BMI at listing, gender, race/ethnicity, diabetes status at listing, and exception status.

Feel free to message me if you have any questions.

Best,
Eric